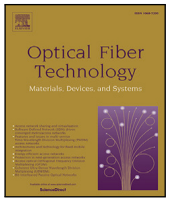
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Regular Articles



Routing and dynamic core allocation with fragmentation optimization in EON-SDM

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ABSTRACT

The increasing need for rapid data transmission in optical networks has made the effective allocation of spectrum resources a crucial necessity. This study presents a new spectrum allocation approach that incorporates path selection and careful consideration of cross-talk (XT) limitations. We propose a novel fragmentation coefficient and dynamic core-changing-based routing, spectrum, and core allocation (FC-DCC-RSCA) technique. The algorithm begins by searching for the shortest paths using the Dijkstra algorithm. It then proceeds to evaluate fragmentation coefficients (FC) using the Continuous Aligned Slot Ratio (CASR) model. Subsequently, the FC undergoes a filtration process wherein only those falling inside predetermined fragmentation ranges (min and maxFC) are retained. Further spectrum allocation has been done based on XT constraints. The algorithm has better proficiency in dynamic flexibility, enabling it to effectively handle varying traffic loads. By conducting a series of rigorous simulations, our proposed approach exhibits a notable decrease in bandwidth blocking probability (BBP) while also opening the door for considerable improvements in spectrum utilization. The algorithm's proficiency in adapting to changing network circumstances signifies its importance as a helpful thing in enhancing the allocation of resources within optical networks. It is well-positioned to address the developing requirements of contemporary communication systems. The spectrum utilization of the FC-DCC-RSCA increases by 12.5% and 6.25%, exhibits a reduction in its FR by 13.5% and 2.7%, and has a BBP of 0.051 for the traffic load of 450 Erlang, compared to SBP-PMC and FMDE-RSCA. The algorithm that has been provided provides opportunities for more investigation and enhancement. Potential areas for future research include the examination of strategies to improve energy efficiency, the development of scalable solutions for bigger network infrastructures, and the incorporation of machine learning methodologies to facilitate dynamic decision-making processes. Furthermore, it is worth investigating the potential of the method to be applied in developing optical communication technologies, such as space-division multiplexing. These endeavors hold the potential to enhance the current level of knowledge and practices in the field of optical network management and optimization.

1. Introduction

In recent years, the elastic optical network (EON) has been intensively researched and is used as a remedy for the inefficient spectrum utilization problem of conventional wavelength division multiplexing (WDM) optical networks [1,2]. In conventional WDM networks, the channel bandwidth is fixed at 50 GHz, which is incompatible with the connection requests of clients with varying bandwidth requirements [3]. For low-data-rate connections where only a small portion of the channel is utilized for the allocation, a 50 GHz channel must be allocated entirely while only a small portion of the channel is utilized for allocation [4].

EON is a highly promising alternative for the establishment of high-capacity networks. In the present day, within the framework of EON, it is possible to partition bandwidth into discrete frequency slots (FS), which facilitate the transmission of data at high rates. When initiating a new connection, the connection specifies its necessary bandwidth by quantifying it in terms of the number of frequency slots (FS). The connection will be accepted and then established if the network can designate the requested FS in a continuous pattern for each link of the selected route. Otherwise, access will be denied [5–7]. EON utilizes subcarriers that possess a significantly narrower frequency slot bandwidth (12.5 GHz) in comparison to traditional fixed-grid WDM networks. The use of a network facilitates the establishment of lightpath connections through the careful selection of an appropriate quantity of closely positioned subcarriers. This approach offers enhanced flexibility

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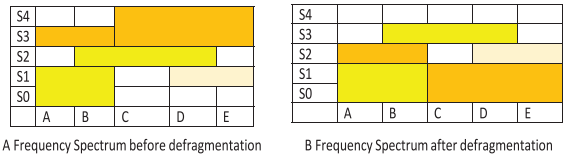


Fig. 1. Frequency spectrum scenario.

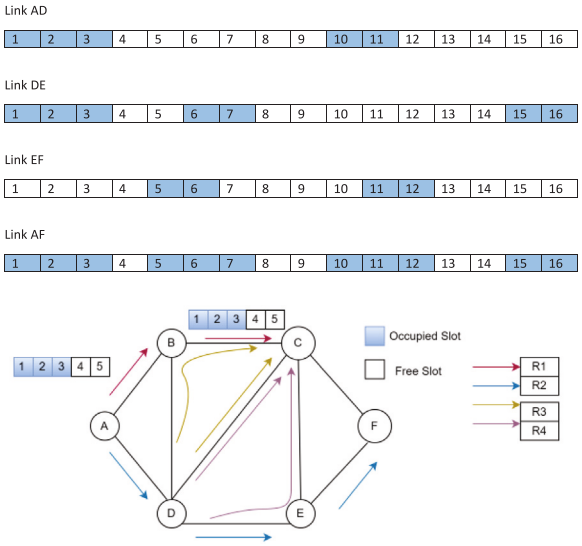


Fig. 2. Link and path fragmentation and spectrum constraint.

and improved efficiency in the utilization of fiber bandwidth [4]. Nevertheless, as a result of the remarkable expansion of network demands and the constraints imposed by nonlinear effects in optical fiber, the transmission capacity of a single-core EON has nearly reached the Shannon limit [8]. The implementation of SDM technology has emerged as a very effective method for enhancing transmission capacity to meet the growing need for network bandwidth. The EON-SDM technology demonstrates a notable capability to adequately address the substantial capacity requirements of modern optical networks. EON-SDM combined with multi-core fiber (MCF) is regarded as one of the most promising transmission technologies for the next generation of optical networks [9]. EON provides flexibility, but it also offers challenges. One of those challenges is to mitigate the fragmentation in an EON [10].

Due to the completion of existing connection requests, the spectrum slots of a link become fragmented, thus creating free slots between the other existing connections. The availability of these unoccupied areas may hinder the formation of future connection requests due to the absence of contiguous spectrum. As the available spectral resources become increasingly limited and fragmented, the ability to make connections within the EON diminishes. In this particular scenario, the spectrum exhibits fragmentation, necessitating a defragmentation procedure to consolidate the available spectrum slot for optimal use.

There are two prevalent strategies for addressing the issue of fragmentation. One approach involves the utilization of fragmentation-aware techniques throughout the routing and spectrum allocation (RSA) procedure for connection establishment. In these methodologies, appropriate routes and available spectrum slots are selected in order to minimize fragmentation inside the network [6,11]. The level of fragmentation is reduced when identical bandwidth requests are assigned to dedicated cores.

As an illustration, let us examine a network comprising five consecutive links denoted as A, B, C, D, and E. Every link is associated with five FSs. Additionally, it is important to evaluate five connections, as seen in Fig. 1. It is evident that, despite the presence of some free FS inside the network, establishing a connection between A and E is unattainable due to spectrum fragmentation. Following the process of defragmentation, the distribution of FS inside the network would resemble the configuration depicted in Fig. 1. This representation illustrates that a link may be created from point A to point E by utilizing a single FS [12].

An example of an implementation within the context of SDM-EON is the integration of a fragmentation-aware RSCA approach. This method incorporates many factors and employs Multiple Factor Decision Making (MFDM) techniques to determine the optimal route, core, and spectrum allocation, with the objective of minimizing the fragmentation rate [12]. Nevertheless, this approach is unable to completely address the issue of fragmentation. Another approach involves employing defragmentation techniques, which aim to consolidate available free spectrum slots by relocating connections to more optimal locations.

Consequently, the network is able to accommodate a greater number of connection requests. The process of selecting an appropriate connection and determining the optimal location for this connection is a complex task. The method given in this paper aims to tackle the problem of resource allocation in EON. EON has become increasingly prominent because of its capacity to effectively adjust to changing traffic needs. Allocating spectrum slices of various sizes achieves this, improving resource utilization. The technique presented in this paper aims to enhance the efficiency of creating light path connections among network nodes by effectively using the available optical spectrum resources.

This introductory section presents a comprehensive outline of the algorithm's objective, emphasizing the significance of CASR and FC while also introducing the dynamic core selection component. The preceding content establishes the foundation for the upcoming parts, where a comprehensive breakdown of the algorithm's phases will be provided. RSA in communication networks must adhere to three essential constraints: spectrum continuity, spectrum consistency, and spectrum non-overlapping. The spectrum continuity restriction refers to the requirement that each link in the transmission line must use FS that is sequentially adjacent in index. The spectrum consistency constraint refers to the requirement that each link in the transmission line must utilize the same FS according to its index. The principle of non-overlapping spectrum constraint refers to the condition where the assigned frequency slots are strictly prohibited from overlapping with one another.

The diagram in Fig. 2 illustrates the path A-B-C, which comprises two links, namely AB and BC. If R_1 requires three FS, it should be allocated to the FS that have consecutive FS indices, such as FS→(1,3). Furthermore, it is necessary to assign link AB and link BC to the identical FS index while ensuring that there is no overlap between them [13]. The routing, spectrum, and core allocation (RSCA) problem must be addressed in EON-SDM to ensure compliance with core constraints. The NP-hard nature of the RSCA issue in EON-SDM has been well acknowledged [14].

In EON-SDM, the establishment and termination of various requests lead to the dynamic establishment and release of lightpaths. Consequently, there is a presence of underutilized and discontinuous fragmented frequency slots in the link [15]. Nevertheless, the use of this fragmented FS for allocating new requests poses challenges. Consequently, the network's spectrum utilization will degrade, ultimately resulting in a substantial increase in the blocking likelihood. The presence of discontinuous idle frames on a connection might result in link fragmentation. In this particular scenario, the allocation of spectrum resources on a singular connection is divided into fragmented and discontinuous segments. As seen in Fig. 2, the path AF comprises the links AD, DE, and EF. However, due to link fragmentation, these links cannot be allocated as a unified big block. However, the presence of discontinuous free frequency slots (FS) on individual links within a path might be seen as a kind of path fragmentation, as it violates

Table 1
Comparison of related papers.

Aspect	H. Oliveira et al. [18]	S. Paira et al. [19]	Xiang et al. [20]	Xuan et al. [21]	Proposed
Fragmentation assessment	Partial	Complete	Complete	Complete	Complete
Adapting core dynamically	No	No	No	No	Yes
Optimized resource assignment	Yes	No	No	Yes	Yes
Delay efficiency	No	Yes	No	No	No
Shared backup path protection	Yes	No	No	Yes	No
Efficient resource utilization	Partial	Yes	Potential for improvement	Yes	Yes
Real time adaptation	No	No	Potential for improvement	Potential for improvement	Yes
Complexity	Moderate to high	Moderate to high	High	High	Moderate to high
Sensitivity	No	No	–	–	Yes
Flexibility to network changes	Low	Low	High	High	High
Exposure of optimization aspects	Partial	Limited	Complete	Complete	Complete

the need for spectrum consistency. According to the data presented in Fig. 2, it is seen that a total of 4 frequency slots (FS) may be allotted for R_2 on each link. However, it is noted that R_2 is unable to be broadcast on Path AF. Consequently, the occurrence of link fragmentation and path fragmentation might result in transmission problems when dealing with high-speed and long-distance requests. According to Jafari et al. the occurrence of link fragmentation has the potential to result in path fragmentation [16]. Consequently, effective control of both components will result in enhanced spectrum efficiency [17]. The fundamental principle of the proposed method fragmentation coefficient and dynamic core changing-based routing and spectrum allocation (FC-DCC-RSCA) is centered on the notion of contiguous aligned slot ratio (CASR) as a crucial measure for assessing the accessibility of contiguous spectrum slots on a certain core. The system incorporates a dynamic core selection mechanism, which enables adaptive resource allocation in situations where the XT restrictions are not initially satisfied.

The technique employed in this context seeks to optimize the efficient usage of the spectrum resources that are already accessible. This is accomplished by employing a parameter known as the FC. Moreover, the system possesses a dynamic component that enhances the network's ability to adapt to different traffic requirements. The model aims to assign the desired spectrum slice (SS) while maintaining compliance with the XT level restriction along the specified path. This step guarantees that the assigned SS does not result in excessive interference or crosstalk with the neighboring slices. If the XT constraint is met, implying if XT is less than $XT_{threshold}$, the SS is assigned. The objective of this study is to offer a thorough manual for the development and deployment of the algorithm in practical EON by conducting a methodical analysis of its functioning. The rest of the paper is as follows: Section 2 summarizes related works, and Section 3 presents the problem statement. Section 4 describes the proposed algorithm. Section 5 gives the simulation results of the proposed algorithm and discusses the performance evaluation results. Finally, Section 6 summarizes the research in the paper.

2. Related work

Methods for RSA and Routing, Modulation, and Spectrum Allocation (RMSA) have been presented in [22] with consideration for the Quality of Transmission (QoT) and the computational complexity. Multipath RSA has been presented in [11,23] in which if an incoming connection cannot be assigned in the designated path, the network attempts to allocate it in alternative paths. Fragmentation Aware with Routing and Spectrum Assignment (FA-RSA) has been proposed in [24] by taking into account external fragmentation measurement. The research presents a novel method that integrates deep learning techniques with a hierarchical graph-assisted strategy to efficiently tackle the problem of crosstalk-aware fragmentation in space division multiplexing elastic optical networks [25].

If any lightpath in the SDM-EON fails, it might result in significant data loss or disruptions in service. Hence, lightpath protection holds significant importance as a subject of research. The concept of network fragmentation is a recent development in spectrally spatially flexible optical networks (SS-FON), but it has been extensively researched in the context of EON [11,16,26–29]. SS-FON is a technique that integrates EON with SDM, leading to enhanced transmission capacity and greater adaptability in the spectrum domain. The increased adaptability in the spatial domain of SS-FON results in the inability to directly apply fragmentation measures, methodologies, and algorithms designed for EON to SS-FON. As far as we know, the measurement of fragmentation in SS-FON has only been examined in a limited number of articles [26]. These papers focus on vertical fragmentation, horizontal fragmentation, or the remaining holding times of dynamic requests. The literature extensively covers the analysis of shared lightpath protection and survivable multipath protection. The study in [13] focuses on the shared backup path protection of EON. However, it only provides a partial assessment of fragmentation and does not include dynamic core adaptability. The CMDE-RSCA (crosstalk aware) algorithm, as described in [30], offers two approaches to enhance fragmented link core XT. However, it does not incorporate dynamic core allocation. Moreover, these literary works lack the incorporation of dynamic core alteration and fail to strike a balance between spectrum utilization and fragmentation. As a result, they lead to poor resource utilization and an increased burden on the network's bandwidth.

The access blocking probability metric [31] can be employed to assess the level of fragmentation. However, like the external fragmentation metric, the access blocking probability measurement is unable to differentiate between three scenarios: (i) all slots are available, (ii) all slots are occupied, and (iii) all available spaces are adjacent to each other [32]. The authors in [31] introduced a fragmentation-aware routing and spectrum allocation system (FA-RSA) that takes into account fragmentation measurement. In this study, the authors examined an algorithm called LBFA (load balancing and fragmentation metric-based fragmentation-aware RSCA) to minimize fragmentation in SDM-EON that enable super-channels (SChs).

The authors in [33] provide a statistic called contiguous accessible resource separation degree (CARSD) to quantify spectrum fragmentation along a given path. It illustrates the correlation between the overall quantity of accessible slots and the dimensions of each continuously available block in every path. Examined the RSA problem related to reservation requests in EON and created a routing approach called minimum resource consumption (MRC) [34]. The authors presented a range of approaches in SDM-EONs, specifically in [19,35], to address the issues of spectrum fragmentation and inter-core crosstalk. The authors presented a range of approaches in SDM-EONs, specifically in [19, 35], to address the issues of spectrum fragmentation and inter-core crosstalk.

The foregoing research lacks a detailed analysis of the fragmentation and sharing of survivable multipaths. Table 1 provides a concise overview of several literature sources that discuss RSCA and its activities in the context of SDM-EON. In order to resolve this issue, we introduce a heuristic technique that utilizes a fragmentation coefficient and incorporates dynamic core switching. The technique maximizes the efficiency of spectrum resource allocation by utilizing the fragmentation coefficient based on CASR under $FC_{threshold}$ value. In addition, the

dynamic core altering element improves the network's ability to adapt to different traffic needs. The Table 1 provides a comprehensive comparison of different algorithms designed to improve resource allocation, including our suggested technique. Each method is evaluated based on crucial parameters such as fragmentation analysis, dynamic core adjustment, prioritization of delay efficiency, and spectrum management, among other criteria. The "FC-DCC-RSCA" algorithm is distinguished by its thorough approach to fragmentation consideration, adaptable modification of the core, and spectrum management. Moreover, it demonstrates the capacity to adjust promptly and efficiently to use resources beyond the constraints of shared backup path protection (SBP). However, it may be susceptible to parameter optimization. FMDE distinguishes itself with its exceptional capacity to promote delay efficiency and minimize fragmentation. Moreover, it guarantees to improve the utilization of resources. The primary goal of SBP PMC is to offer shared backup path protection for networks that have many cores. The methodology proposed by Xiong [25] combines deep learning with a hierarchical graph-based strategy to address crosstalk and prevent fragmentation. It uses deep learning to gain an understanding of complex patterns in network data. The hierarchical graph enables the systematic allocation of resources, efficiently addressing issues associated with crosstalk and fragmentation. The [36] study primarily examines the safeguarding of resources in elastic optical networks that employ space division multiplexing and undergo fluctuating traffic. It ensures the protection of important connections amid changing network conditions. The study undertaken by Zhao et al. focuses on the process of crosstalk-aware spectrum defragmentation in multi-core fiber networks through the re-provisioning of advance reservation requests [37,38].

A review of resource allocation algorithms and schemes in a flexible spectral–spatial optical network is presented in [24]. Two algorithms are presented in [39,40] to enhance QoS parameters such as the average number of calls in queue and the probability of blockage. A method for routing, spectrum assignment, and core allocation in eon with multiple cores and maximum index used frequency slots is described in [21]. In [16], the Fragmentation-Aware Best Splits (FABS) algorithm employs fragmentation criteria, and the Best Split and Crosstalk Aware Best Split (CABS) algorithm as a novel split technique. These two algorithms can increase spectrum utilization and decrease obstruction probability. CMDE-RSCA (crosstalk-aware) algorithm and FMDE-RSCA (fragmentation- aware) algorithm are two methods presented in [19] for improving fragmentation and inter-core crosstalk in EON-SDM.

To reduce obstruction probability and fragmentation in EON-SDM, fragmentation-aware and time-based algorithms based on holding time and variable CVM coefficient have been introduced in [26]. In [35], methods for reducing interference and blockage probability in EON-SDM have been introduced by taking the physical layer's security level into account. A method for routing, spectrum assignment, and core allocation based on Multiple Criteria Decision Making (MCDM) methods in eon with multiple cores, where cores are classified based on the holding time of an input request, is presented in [12]. Traffic Awareness Crosstalk Interference Avoidance (TACIA-RMSCA) is a Routing, Modulation, Spectrum, and Core Allocation (RMSCA) algorithm that overcomes inter-core crosstalk (XT) and calculates the performance of MCF-SDM-EON for peak load of the fluctuating traffic, as presented by the authors in [37]. The primary objective of [38] is to investigate time-spectrum defragmentation in EON-SDM. This paper presents the Time-dimensional Spectrum Compactness (TSC) metric. A Crosstalk-aware Re-Provisioning (CRP) algorithm with two strategies to re-provision Advance Reservation AR requests and increase spectrum efficiency in EON-SDM has been developed using this metric. Routing, Modulation, Spectrum, and Core Allocation (RMSCA) algorithm Distance-adaptive Energy-aware Resource Allocation (DERA) has been presented in [36] to reduce the dynamic XT effects in EON-SDM by taking into account the survival multipath scheme and Spectrum Compactness (SC) during resource allocation. A Routing, Modulation, and Spectrum Assignment (RMSA) method in elastic optical networks has been presented in [41]. This method chooses an appropriate block for each request that has the smallest aggregate of weighted resource reductions. Considering core classification in EON-SDM, the research in [42] reduces spectrum fragmentation. In [9,43], a method is presented for classifying the cores based on the number of desired frequency slots store. A common core is the final core that is sought for allocation in this task. Note that any number of frequency positions can be allocated for any request in the C_{th} core. The (C-1)th core is organized for the region of one FS, whereas the other cores are categorized according to their prime numbers. When the same bandwidth requests are allocated to specific cores, fragmentation decreases.

Spectrum Block Multipath per Core (SBMC) is a spectrum and core classification method presented in [23] for EON-SDM under MCF. Using a multipath mechanism and factoring in core classification, this method reduces the likelihood of blockage. The KSP-FASA routing and spectrum allocation algorithm was introduced in [44] in an effort to reduce spectrum fragmentation and increase spectrum efficiency. In this algorithm, a block cost function is specified for evaluating spectrum candidates along a given path. After spectrum allocation, the block cost function is investigated based on the status of adjacent frequency slots in order to reduce fragmentation.

The First Core Fit (FCF) algorithm [26] is a method of spectrum allocation and routing for EON-SDM under MCF, that selects the first empty block appropriate to the request for allocation and does not consider fragmentation. In the study conducted by Paira et al. (2020), two algorithms were introduced, namely CMDE-RSCA (crosstalk-aware) and FMDE-RSCA (fragmentation-aware [19]). The fragmentation-aware algorithm demonstrated the capacity to significantly decrease the likelihood of bandwidth blockage. This finding suggests that the fragmentation-aware algorithm is useful in mitigating bandwidth limitations.

In the study conducted by Oliveira et al. (2019), three routing and spectrum allocation (RSCA) methods were introduced. These algorithms were specifically designed to address the issue of single-link failure in EON-SDM [18]. All three algorithms employed the RSA technique, which is based on a multi-graph representation of the spectrum. Additionally, all three strategies utilized the shared backup path approach. Among the three RSCA algorithms considered, it can be observed that the shared backup path protection for multicore network algorithm (SBPPMC) has superior performance.

3. Problem statement

Spectrum allocation for a specific connection request is a crucial challenge in the EON. Traditional RSA algorithms often fail to achieve an equilibrium between spectrum utilization and fragmentation. This results in suboptimal resource utilization and an increased blocking probability. In order to resolve this issue, we propose a heuristic method that uses the CASR as a fragmentation metric. The algorithm seeks to identify appropriate paths and cores for every request, optimizing spectrum allocation while adhering to XT-level constraints. The objective is to increase spectrum utilization and decrease blocking probabilities in EON. This research centers on comparing the performance of the proposed heuristic to that of conventional RSA methods, evaluating its efficacy in dynamic network scenarios, and determining its influence on network efficiency metrics.

The efficient allocation of spectrum in EON presents a formidable obstacle. Traditional RSA algorithms struggle to achieve a satisfactory equilibrium between spectrum utilization and fragmentation reduction. As a consequence, there is inefficient utilization of resources and an increased likelihood of obstruction. In order to resolve this issue, we propose a heuristic strategy that employs the CASR as a fragmentation metric. The algorithm is intended to identify optimal paths and cores for each request, optimize spectrum allocation, and ensure compliance with XT-level constraints. Its objective is to maximize spectrum utilization and reduce the likelihood of obstruction.

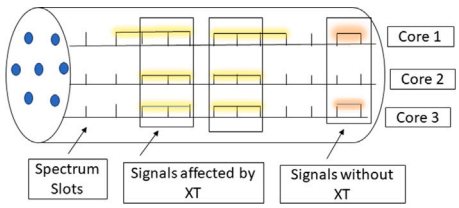


Fig. 3. InterCore crosstalk.

The ability to allocate the spectrum with precise granularity allows for efficient management of diverse traffic demands. Nevertheless, the utilization of several cores results in XT, which adversely affects the transmission quality and diminishes its availability of the spectrum for future allocations. XT refers to the interference caused by adjacent cores on the same optical fiber link. The term refers to the increase in optical power resulting from the contribution of neighboring links to the power of a signal in a core, divided by the power in that core, and expressed in decibels (dB).

From the perspective of core and spectrum resource allocation, inter-core XT only affects the lightpath that occupies the same spectrum position in adjacent cores at the same time [32]. Since XT originates from the evanescent wave between adjacent cores, it only affects the lightpaths in adjacent cores, while the influence between non-adjacent cores is negligible [33]. For instance, as shown in Fig. 3, if spectrum slots are overlapping on all the three cores then they create a XT interference. Signals affected by the XT and the signals which are not affected by XT is shown in the Fig. 3. RSCA algorithms must ensure that the allotted spectrum for a connection meets the necessary conditions for the signal to be successfully decoded at the destination. Hence, it is necessary to verify the extent of crosstalk in a specific spectrum region before assigning it to a link. If the value exceeds an unacceptable threshold, the region should not be allotted. Furthermore, even if the crosstalk level is below the threshold value, it is necessary to evaluate the impact of the incoming connection's crosstalk on the crosstalk of all other connections. If the level of crosstalk exceeds the predetermined limit, the connection must be rejected, as it will result in a deterioration of transmission quality for existing connections. This study centers on a comprehensive evaluation of the proposed heuristic in comparison to conventional RSA methods, assessing its efficacy in dynamic network scenarios, and determining its impact on network efficiency metrics. The selection of FMDE-RSCA as a benchmark approach is based on its suitability for addressing two significant issues, namely inter-core crosstalk and fragmentation inside the core. Previous research or studies may have been conducted to establish the efficacy of FMDE-RSCA in mitigating inter-core crosstalk and fragmentation concerns. This approach would be considered appropriate as a baseline for comparative analysis. This suggests that our study is primarily focused on addressing these challenges in order to ensure high-quality signal transmission. The use of SBPPMC as a benchmark methodology is justified due to its specialized design for multi-core networks. This suggests that our study is centered around this particular network design, and SBPPMC serves as a suitable benchmark given its compatibility with the context of our research. The SBPPMC places considerable importance on the utilization of shared backup pathways as a means to safeguard primary paths. This function effectively tackles the crucial matter of guaranteeing network resilience and fault tolerance, which is a prominent worry in several network architectures. The implementation of SBPPMC has the potential to achieve a harmonious equilibrium between the preservation of network performance, achieved via the provision of backup route protection, and the optimal utilization of available resources. Proper distribution and management of spectrum resources is a critical factor to be taken into account.

3.1. Network model

EON-SDM is represented by a network topology-based connectivity graph G (V, E, C, SS), in which V represents the number of nodes and E represents the fiber link set. C represents the collection of cores on a link, and its notation is $C = C_1, C_2, \ldots, C_n$. Each fiber core in a link carries a certain number of frequency slots, or SS. We have implemented the proposed algorithm using MATLAB. Initially, a network graph is constructed. Following this, an adjacency matrix, which functions as a binary map indicating the connections between these nodes, is defined. A value of 1 in the matrix signifies a connection, while 0 denotes no direct link. Connection Requests: A new connection request (NCR) is denoted as (S, D, SS), where S represents the source node, D represents the destination node, and SS represents the requested spectrum slice. This indicates a request to establish a light path between two nodes with a specified spectrum requirement.

Path Selection (Algorithm 1): The algorithm begins by using the Dijkstra algorithm to identify the k-shortest paths between the source and destination nodes. These paths depict possible connections for the light path.

Core Selection (Algorithm 2): After determining the k-shortest paths, the algorithm selects a suitable core for the connection. Based on CASR, the FC of each path is evaluated. CASR assesses the availability of spectrum channels that are contiguous on a core. Paths that do not satisfy the predefined FC constraints (minFC and maxFC) are discarded, and the remaining paths are classified based on their FC values. The path with the lowest FC value and its associated core are chosen.

Spectrum Slice Assignment with XT Constraint and Dynamic Core Selection (Algorithm 3): The model attempts to allocate the requested spectrum slice (SS) while adhering to the XT (crosstalk) level constraint with the given path. This step ensures that the allotted SS does not cause excessive interference or crosstalk with the adjacent slices. If the XT constraint is satisfied, that is, if $XT < XT_{threshold}$, the SS is allocated; otherwise, the algorithm seeks iteratively for an accessible SS by evaluating alternative paths and cores as needed. If the XT constraint is not satisfied, the model includes a provision for dynamically altering the core to allocate spectrum using alternative cores. This adaptability permits the network to accommodate varying conditions and constraints, ensuring successful SS allocation while minimizing crosstalk. The value is returned as the designated path (k) and core (c), which indicate the beginning and ending segment indices of the allocated spectrum block. This information is essential for establishing a link. Eq. (1) represents the quantity of spectrum slots necessary for a specific connection request. N denotes the number of necessary contiguous spectrum slots. B_{GB} is a symbol for the guard band capacity. R denotes the bandwidth request. C'_{slot} represents the capacity of the slot, and generally it is 12.5 GHz. m represents the modulation type.

$$\lceil N \rceil = \frac{R}{C'_{slot}.m} + B_{GB} \tag{1}$$

The Eq. (1) represents the computation required to determine the number of spectrum slots required for a certain connection request in a communication system. The equation considers multiple parameters to determine the necessary allotment of bandwidth. The equation considers multiple parameters to determine the necessary allotment of bandwidth. N is the total number of consecutive spectrum slots that are required. The equation is used to calculate this quantity. B_{GB} refers to the guard band. These are employed to create separation between channels in order to mitigate interference. C_{slot} is the slot capacity it is the maximum amount that a single slot can hold designated as 12.5 GHz, this represents the capacity of bandwidth that may be contained in a single slot. m denotes the type of modulation. Modulation techniques exhibit diverse bandwidth requirements and efficiency. This information is vital for providing the requisite resources

for communication networks to guarantee efficient and uninterrupted data transfer.

The objective of this equation is to ascertain the quantity of consecutive spectrum slots needed to meet a specified bandwidth need, taking into account parameters such as guard bands, slot capacities, and modulation type. This information is vital for providing the requisite resources for communication networks to guarantee efficient and uninterrupted data transfer. EON-SDM introduced space division multiplexing technology, which significantly increases network capacity but also has an inevitable impact, namely the impact of XT. Crosstalk between adjacent cores is a crucial concern in MCF because it degrades transmission quality [37].

XT refers to the disruption caused by neighboring cores on a shared optical fiber network. The term refers to the decibel (dB)-expressed amplification of optical power due to adjacent connections in relation to the power of a signal in a core. The figure illustrates the XT in multicore fiber. RSCA algorithms must ensure that the allotted spectrum for a connection meets the necessary conditions for the signal to be successfully decoded at the destination. Hence, it is necessary to verify the extent of crosstalk in a specific spectrum region prior to assigning it to a link. If the value exceeds an unacceptable threshold, the region should not be allotted. Furthermore, even if the crosstalk level is below the threshold value, it is necessary to evaluate the impact of the incoming connection's crosstalk on the crosstalk of all other connections. If the level of crosstalk exceeds the predetermined limit, the connection must be rejected, since it will result in a deterioration of transmission quality in existing connections. Eq. (2) is utilized to determine the XT between one core and n neighboring cores in a homogeneous MCF fiber.

$$XT = \frac{n - n * (-(n+1) * h * L)}{1 + n * (-(n+1) * h * L)} \tag{2}$$

where h is calculated by Eq. (3).

$$h = \frac{2.k^2.R}{\beta.\ D} \tag{3}$$

where:
h is crosstalk increase per unit length;
k is the coupling coefficient;
β propagation constant;
R bend radius of fiber;
D core-pitch (distance between cores)
L is the length of the fiber;
n is the number of adjacent cores;

In this paper, we make the same realistic assumptions assumed in [45] as $k = 2 \times 10^{-5}$, R = 50 mm, $\beta = 4 \times 10^6$, D = 45 μm.

The Eq. (3) provided for determining crosstalk (XT) between one core and n neighboring cores in a homogeneous MCF pertains to the effects of linear impairments in optical fiber communication systems.

4. Fragmentation coefficient routing, core and spectrum allocation with dynamic core

In this section, we present a fragmentation coefficient-based heuristic algorithm with dynamic core switching. The strategy optimizes the allocation of spectrum resources by utilizing the fragmentation coefficient. Additionally, the dynamic core-altering aspect enhances the adaptability of the network to varying traffic demands. Upon receiving a new connection request (NCR), the algorithm identifies the k-shortest paths or routes between the source and destination nodes. The fragmentation coefficient ($FC^{sd}_{k,c}$) is then computed for each combination of the k_{th} route and c_{th} core using CASR. Then, these coefficients are used to filter the pairs, discarding those whose $FC^{sd}_{k,c}$ values fall below min(FC) or exceed the max(FC). The remaining pairs are sorted in ascending order, and the pair with the smallest min (k,c) and its relative FC value is chosen. This flowchart depicts the process of our proposed technique as FC-DCC-RSCA as shown in Fig. 4.

4.1. Modified heuristic approach with minFC, maxFC and dynamic core changing

The initial phase of the algorithm, known as "path selection", plays a pivotal role in route selection for a newly requested connection. The initial step of this phase involves the inclusion of three essential inputs: the source node (S), which denotes the origin of data transmission; the destination node (D), which indicates the final destination for data delivery; and the requested spectrum slice (SS), which outlines the specific frequency range or slots required for the transmission. Subsequently, the method continues to compute the k-shortest paths between the source node (S) and the destination node (D). The 'k-shortest path' concept in Algorithm 2 makes it easier to take into account a variety of options, adding flexibility to path selection. The mentioned routes serve as the fundamental basis, facilitating the determination of the optimal route and desired spectrum slice. The first part of this process is crucial in determining a variety of feasible choices for the progression of data inside the network, which depends on distinct paths.

Algorithm1 Heuristic Approach

Input: Light path request.
Output: Spectrum allocation of the light path.
1: Whenever a novel connection request, denoted as NCR (S, D, SS), is received, the objective is to determine the k-shortest paths or routes connecting the source and the destination.
2: Find the fragmentation coefficient ($FC^{sd}_{k,c}$) based on CASR for each kth and c^{th} pair.
3: Filter the list of k-c pairs to include only those pairs where the FC is within the range [minFC, maxFC].
4: Sort the filtered k-c pairs in ascending order of FC and choose the pair with the minimum FC value. Let's denote this pair as (k_{min},c_{min}).
5: Determine the suitable SS value while maintaining adherence to the acceptable degree of XT constraint by using the path and core of the chosen pair (k_{min},c_{min}).
6: If SS is not accessible or does not meet the XT level constraint: Take the pair (k_{min},c_{min}) off the filtered list. Return to step 4 and choose the pair with the next-lowest FC value. Repeat this procedure until an appropriate spectrum slot is identified or the filtered list is vacant.
7: If a spectrum position is accessible, return the route k_{min} and core c_{min} (beginning slice index and ending slice index) of the selected spectrum block.

Algorithm 2 Path Selection

Input: Source Node (S), Destination Node (D), Requested Spectrum Slice (SS).
Output: List of k-shortest paths.
1: Determine k shortest paths or routes from the source to the destination.
2: Provide a list of k shortest paths.

Algorithm 3 Core Selection

Input: List of k-shortest paths, minimum fragmentation coefficient (minKC), maximum fragmentation coefficient (maxFC).
Output: Selected Core (C) and corresponding FC value.
1: For every path in the collection of k-shortest paths, the fragmentation coefficient ($FC^{sd}_{k,c}$) is computed using the CASR method.
2: If the value of $FC^{sd}_{k,c}$ is less than the minimum value of FC or more than the maximum value of FC, the pair should be eliminated.
3: Sort the rest of the pairs by ($FC^{sd}_{k,c}$) and put them in increasing order.
4: Pick the pair where min(k,c) is the lowest.
5: Given back the chosen core (c) and the FC_{sd} value.

The objective of considering max(FC) is to calculate the number of consecutive spectrum slots needed to meet a specific bandwidth need, taking into account parameters such as guard bands, slot capacities, and modulation type. This information is vital for providing the requisite resources for communication networks to guarantee efficient and uninterrupted data transfer.

Through the examination of FC values for different maxFC values, it is possible to determine the range of thresholds that result in satisfactory or optimal resource allocation. By examining the FC values for different maxFC values, one can determine the range of thresholds that result in satisfactory or optimal resource allocation. This research helps determine the maximum level of fragmentation that the system can tolerate while maintaining optimal resource utilization. The highest FC value serves as a standard for assessing the effectiveness of resource allocation algorithms. By comparing the achieved FC values for different maximum FC values, we may evaluate the efficiency of alternative resource allocation strategies. This research can aid in selecting the most suitable allocation method that efficiently balances fragmentation and other system performance criteria. By analyzing the highest FC value throughout the investigation and studying FC values, we may gain useful insights into how fragmentation thresholds impact the effectiveness of resource allocation. This information can offer advice for the decision-making process, optimize resource allocation, and boost the overall operation of the communication system. Algorithm 3, referred to as "core selection", plays a pivotal role in the determination of the most appropriate core for the purpose of data transmission. The input for this phase consists of a list of k-shortest paths as well as the designated minFC and maxFC fragmentation coefficients. The approach computes the fragmentation coefficient ($FC_{k,c}^{sd}$) for each path in the list of k-shortest paths, utilizing the contiguous aligned slot ratio (CASR) as the basis for calculation. If the computed $FC_{k,c}^{sd}$ falls below the minimum fragmentation coefficient (minFC) or beyond the maximum fragmentation coefficient (maxFC), the pair is eliminated. This phase is essential in order to limit the assessment process to only those core-path pairs that are deemed feasible.

Subsequently, the remaining pairs are arranged in ascending order according to their $FC_{k,c}^{sd}$ values. This enables the determination of the pair exhibiting the lowest $FC_{k,c}^{sd}$, which signifies the core-path combination that provides the highest level of slot contiguity. Subsequently, the algorithm proceeds to choose the pair that possesses the lowest value between min (k,c), indicating the optimal alignment achieved by combining the core and path. The act of selecting is crucial for optimizing the allocation of resources in data transmission.

Ultimately, the outcome of this phase encompasses the chosen core (c) and its accompanying FC_{sd} metric, which offers significant insights for the remaining stages of the algorithm. The mentioned step assumes a crucial function in guaranteeing the suitability of the selected core for the particular source, destination, and desired spectrum slice, hence leading to the optimization of data transmission efficiency and effectiveness within the network.

Algorithm 4 Spectrum Allocation with XT Constraint

Input: Selected Core (C), chosen paths (k), Requested spectrum slice (SS), XT Threshold

Output: Return the allocates SS

1: Determine the appropriate SS which satisfies the XT level constraints, considering the designated core (c) and selected route (k).

2: If SS is inaccessible (XT constraint violated), return a flag indicating allocation failure else, return the allocated SS.

The Algorithm 4 then searches for a suitable SS while adhering to the XT level constraint that is checking the $XT_{threshold}$ every time. It checks whether the XT exhibited by the SS is $< XT_{threshold}$. If it is less than the threshold, then SS is accessible. In the event that the SS is inaccessible, the (k,c) pair is eliminated, and the process continues to search for the best FC pair. If, following these iterations, no suitable pair is identified, the algorithm employs dynamic core switching until a valid spectrum slice allocation is attained. The algorithm then returns the chosen route (k) and core (c), along with the beginning and ending slice indices of the allocated spectrum block, if the SS is accessible. This method optimizes the spectrum allocation procedure by achieving an optimal balance between resource utilization and XT constraints. By intelligently adjusting parameters or strategies, the heuristic can avoid unnecessary computations, potentially leading to faster convergence or termination.

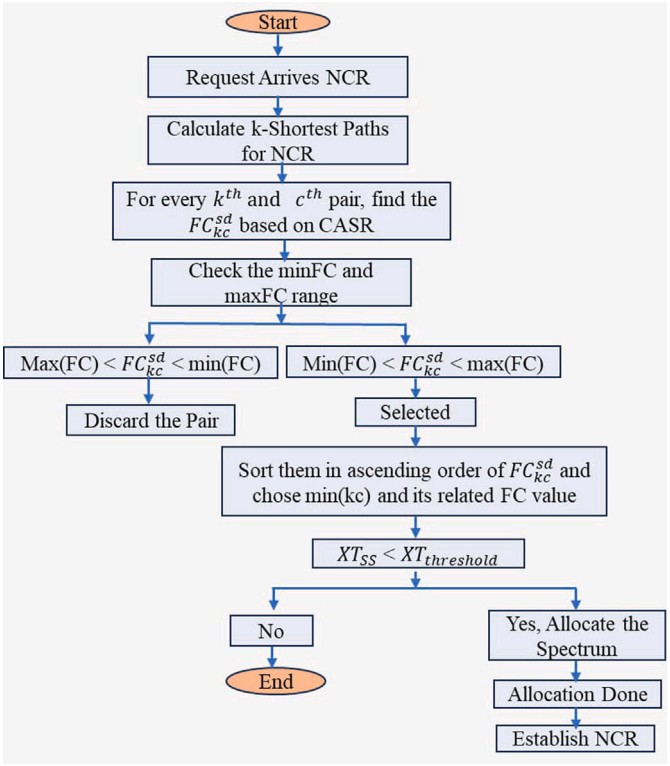


Fig. 4. Flowchart of FC-DCC-RSCA.

Algorithms 2, 3, and 4 are essentially components of Algorithm 1. Algorithm 1 incorporates these three algorithms. Algorithm 2 involves the use of the Dijkstra Algorithm to determine the k-shortest path. Algorithm 3 focuses on the process of core selection. It takes inputs such as paths, minFC, and maxFC values and generates the output of the selected core and its related FC value. Algorithm 4 describes the process of spectrum allocation with the XT constraint. It takes inputs such as the selected core, a specific path, the required SS (spectrum slice), and the XT threshold. The output of the algorithm is the allotted SS.

Dynamic core changes can make the heuristic more robust against changes in problem conditions or characteristics. This can lead to consistent performance across a wider range of spectrum utilization. All three of these algorithms work together to choose paths, cores based on minimum and maximum FC constraints, and spectrum segments that are assigned with XT constraints. In summation, Algorithm 4 is essential for ensuring that the allocated spectrum slice is compatible with the selected core, path, and requested spectrum slice while maintaining acceptable XT levels. This ensures the efficient and trustworthy transmission of data within an elastic optical network.

The technique enhances the allocation of spectrum resources by utilizing the FC. Furthermore, the ability to modify the core dynamically improves the network's flexibility in response to different traffic requirements. Let us understand it by an example. Suppose we have



Fig. 5. FC for path 1.

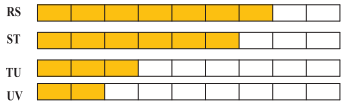


Fig. 6. FC for path 2.

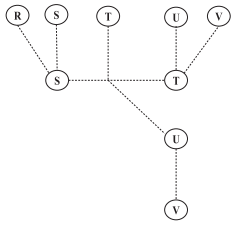


Fig. 7. Topology.

a random topology as shown in Fig. 7 consisting of nodes R,S,T,U,V in which a request arrives from R-V for a required spectrum slot (RSS) of 7 units. Now for the request from R-V with RSS = 7, we find two shortest paths.

Path 1: R-S-T-U-V (Total Spectrum Slots Available = 9)

Path 2: R-T-U-V (Total Spectrum Slots Available = 8)

Now let there is a $FC_{threshold}$ as (minFC = 0.2, maxFC = 1.4), we calculate FC values as according to the notation $FC^{sd}_{k,c}$ as shown in the above figure, first calculating the FC for path 1 and then FC for path 2 as according to Figs. 5 and 6.

1. $FC^{RS}_{1,1} = 1 - \frac{7}{9} = 0.2$
2. $FC^{ST}_{1,1} = 1 - \frac{6}{9} = 0.3$
3. $FC^{TU}_{1,1} = 1 - \frac{3}{9} = 0.6$
4. $FC^{UV}_{1,1} = 1 - \frac{2}{9} = 0.7$

Total $FC^{RV}_{1,1} = FC^{RS}_{1,1} + FC^{ST}_{1,1} + FC^{TU}_{1,1} + FC^{UV}_{1,1} = 1.8$ which does not lies in between the range of minFC and maxFC.

Now calculating the FC for the second path(R-T-U-V) as

1. $FC^{RT}_{2,1} = 1 - \frac{6}{8} = 0.2$
2. $FC^{TU}_{2,1} = 1 - \frac{5}{8} = 0.3$
3. $FC^{UV}_{2,1} = 1 - \frac{3}{8} = 0.6$

Total $FC^{RV}_{2,1} = 1 - \frac{6}{8} + 1 - \frac{5}{8} + 1 - \frac{3}{8} = 1.1$ which lies in the range of minFC and maxFC. Let say the $XT_{threshold} = 0.15$. Calculating the XT value for both the paths using Eq. (2) as $XT_{RSTUV} = 0.16$ and $XT_{RTUV} = 0.13$ so path RTUV is selected else if both the paths do not satisfy threshold condition then dynamic core changing criteria comes into picture.

4.2. Time complexity analysis

1. Path Calculation: The time complexity for finding k-shortest paths in a network depends on the specific algorithm used for the path calculation. The algorithm used is Dijkstra's algorithm. In Dijkstra algorithm, with an adjacency matrix the time complexity is $O(V^2)$ where V is the number of nodes in the graph. With an adjacency list and a binary heap the time complexity is $O(V + E) * \log(V)$, where V is the no. of vertices and E is the no. of edges. The overall time complexity for the fragmentation coefficient calculation is determined by the loop that iterates through the available slots and the computational k-shortest paths. So overall time complexity can be approximated as by Eq. (4).

$$O(totalSlots + (V + E) * log(V)) \tag{4}$$

Assuming the time complexity of the 'CalculateCASR' function is denoted by T (input Size) and let us assume that the input size is determined by the maximum values of totalSlots, route, core and spectrum block. Denote these maximum sizes as S,R,C and B respectively. The outer loop runs M times where M is the length of 'minFC' values. The inner loop runs N times where N is the length of 'maxFC' values. Inside the nested loop the 'calculateCASR' function is called with the following parameters. CalculateCASR(route, core, spectrumBlock, totalSlots, minFC, maxFC) where S,R,C and B represent the maximum sizes of the inputs. Therefore, worst time complexity is depicted by Eq. (5).

$$TC_{worstcase} = O(M.N.T(S, R, C, B)) \tag{5}$$

5. Simulation and analysis

5.1. Simulation parameters

In order to enhance the credibility and applicability of the proposed approach, we conducted a simulation utilizing the NSFNET and USNET network architecture as shown in Figs. 8 and 10. NSFNET architecture is a 14 node architecture while USNET is a 24 node network architecture. NSFNET and USNET seemed very important in setting standards, protocols, and routing methods that have shaped the design of the internet. They played a big role in making the rules and standards for managing networks and transmitting data. They showed how network designs change over time and how important it is for networks to be able to grow, connect, and send data quickly. The requests are received in a random manner, and the corresponding system parameters are presented in Table 2 are described below.

Number of Requests: The number of requests denotes the volume of connection requests that the network must manage. FC-DCC-RSCA efficiently handles these requests by optimizing the routing, selecting the core, and allocating the spectrum. It guarantees the effective utilization of existing resources to satisfy these demands while minimizing the likelihood of blockage and maximizing the performance of the network.

Traffic Load Request: The fluctuating traffic load requests, ranging from 150 to 700 Gbps, indicate changes in the need for network traffic. FC-DCC-RSCA responds to these changes by dynamically modifying spectrum allocation, core selection, and routing to optimize resource utilization and reduce fragmentation, thus ensuring effective management of various traffic loads.

No. of Cores/fiber: The efficacy of FC-DCC-RSCA in regulating the number of cores per fiber is essential for optimizing resource allocation. The system dynamically allocates the 6 cores per fiber based on traffic demands, with the goal of minimizing crosstalk and maximizing the utilization of available cores to accommodate different connection requests.

Number of Frequency Slot (FS's/Link): The quantity of frequency slots per link determines the level of detail in spectrum allocation. FC-DCC-RSCA employs this data to allocate spectrum in a dynamic manner, guaranteeing optimal utilization of frequency slots across connections. It optimizes the allocation of spectrum to prevent fragmentation and maximize the amount of accessible spectrum for meeting connection requests.

Request Demand: It refers to the amount of bandwidth needed for each connection request. FC-DCC-RSCA efficiently handles a wide range of request demands, varying from 10 to 500 Gbps. It maximizes the distribution of available frequencies and the selection of central resources

Table 2
System parameters for a physical layer model.

No. of requests	10^4
Traffic load request	150–700 Gbps
No. of cores per fiber	6
No. of FS's/link	320
Request demand	(10–500) Gbps
Crosstalk threshold ($XT_{threshold}$)	−16 dB

to effectively handle different bandwidth needs, reducing the chances of network congestion and improving overall network performance. $XT_{threshold}$: It is set at −16 dB. It refers to the interference between optical signals, and it is essential to keep it below this threshold to minimize any damage to the signal quality. FC-DCC-RSCA optimizes signal quality and network performance by dynamically adjusting spectrum allocation and routing to keep crosstalk below the set threshold.

FC-DCC-RSCA utilizes these characteristics to assign resources dynamically, optimize spectrum utilization, and efficiently handle various traffic demands within the elastic optical network. This approach guarantees good performance and minimizes the likelihood of blocking. The computation of the simulation indicators involves the use of Eq. (4) for the calculation of BBP and Eq. (8) for the calculation of spectrum utilization (SU). The fragmentation ratio is defined as the proportion of successive free spectrum blocks to the total number of available free frequency slots. The issue in EON that impacts BBP is fragmentation, for which several solutions have been proposed. The fragmentation in the spectra formula is one of the proposed solutions. We employed external fragmentation as one of these factors in this work. One of the enhancements and advantages of our method is the parameter of external fragmentation, which we take into consideration in our suggested strategies. The external fragmentation parameter is employed to decrease the blocking probability in various scenarios and is used to identify fragmentation in spectra [29]. A metric used to illustrate the extent of spreading gaps in spectra is called external fragmentation. A frequency block refers to a collection of adjacent empty spectra. When the allotted bandwidth is partitioned into several tiny contiguous frequency slots that are not in use, the value of F_{ext} approaches 1. Alternatively, in cases where the value of (F_{ext}) approaches zero, the available frequency slots tend to aggregate in a concentrated manner into a certain region of the spectrum. In the context of a particular spectrum, it is crucial to address the utilization ones and zeros, which represent occupied and unoccupied frequency slots, respectively. For example, with 30 frequency slots, consider S_i and S_j as

S_i = (11001001101010111001100000010)→ F_{ext} = 0.625

S_j = (11111111111111001001100000000)→ F_{ext} = 0.3076

According to the percentage of F_{ext} displayed above, S_j is considered to be less fragmented. Eq. (7), which is first presented in [30], quantifies parameter F_{ext} (external fragmentation) = 1 − CASR; where CASR is given through Eq. (6).

$$CASR = \frac{largest\ frequency\ block}{No.\ of\ total\ free\ frequency\ slots} \tag{6}$$

$$F_{ext} = 1 - \frac{largest\ frequency\ block}{No.\ of\ total\ free\ frequency\ slots} \tag{7}$$

$$BBP = \frac{B_{block}}{B_{total}} \tag{8}$$

Eq. (8) comprises the variables B_{block}, which represents the bandwidth allocated to blocked requests, and B_{total}, which represents the aggregate bandwidth of all transmitted requests.

$$SU = \frac{NUS}{TUS} \tag{9}$$

Eq. (9) presents the SU parameter, where by NUS represents the number of used slots and TUS is the total number of slots. We model the performance of the suggested method as well as SBP-PMC and FMDE-RSCA, the comparative algorithms. For a better understanding of the theorem, we computed several parameters as follows: BBP, FC, SU, $XT_{threshold}$ versus spectrum utilization (SU).

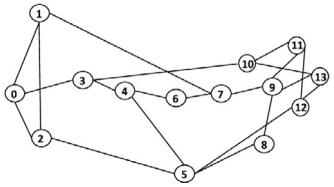


Fig. 8. NSFNET topology.

5.2. Simulation results

The simulation has been carried out for two network architecture NSFNET and USNET as shown in Figs. 8 and 10. The trends indicate that as the traffic load increases, BBP of the two algorithms increases as well. The FC-DCC-RSCA algorithm outperforms SBP-PMC and FMDE due to its multifaceted optimization strategy for EON. FC-DCC-RSCA is unique in that it addresses three essential components simultaneously: routing, core selection, and spectrum allocation. The unique emphasis on the FC and the objective of reducing spectrum fragmentation, which has a significant impact on network efficiency, strengthen this strategy. In addition, FC-DCC-RSCA has a feature called "dynamic core-changing", which lets it change how cores are allocated automatically when network conditions change. Implementing this complete strategy leads to a significant decrease in BBP, hence optimizing network performance through efficient resource utilization and raising the quality of service for consumers.

FC-DCC-RSCA outperforms SBP-PMC and FMDE algorithms in minimizing BBP in by prioritizing optimization of routing, core selection, and spectrum allocation, while also addressing fragmentation and integrating dynamic core-changing capabilities. The holistic optimization technique is designed to improve network efficiency, reduce BBP, and optimize resource utilization, ultimately enhancing the overall performance and quality of service. The offered algorithm permits the dynamic allocation of cores. If no suitable spectrum slice is discovered initially, the algorithm iteratively considers various cores until a valid allocation is found. This flexibility can enhance spectrum allocation success in complex network scenarios. It is possible that SBP-PMC does not offer the same level of flexibility in core selection, which could make spectrum allocation more likely when network conditions are tough.

USNET offers advanced technology and services that are specifically tailored to satisfy the needs of optical networking. The advantages include fast data transfer, low latency connections, scalability, and support for data-intensive applications such as high-definition video streaming, cloud services, and real-time data processing. Consumers looking for flexible and affordable solutions will find the network's cost-effectiveness and scalability attractive because of its ability to offer flexible plans that can accommodate various bandwidth requirements. The advantage of BBP within the USNET network, utilizing the fragmentation coefficient and dynamic core changing, over other algorithms like SBP-PMC and FMDE, arises from essential operational factors such as USNET incorporates the fragmentation coefficient into its algorithm to enhance spectrum utilization by addressing fragmentation problems.

This strategy minimizes the inefficient use of radio frequencies by effectively distributing resources and decreasing the likelihood of connection disruptions caused by a lack of available frequencies, leading to a reduced bandwidth bottleneck. In addition, the dynamic core-changing feature in USNET improves the ability to adjust to changing network conditions. This feature allows for quick adjustments in core allocations, ensuring that resources are used optimally in response to immediate demands. USNET reduces the danger of connection blockages caused by core unavailability by dynamically regulating core allocations, resulting in a decreased BBP.

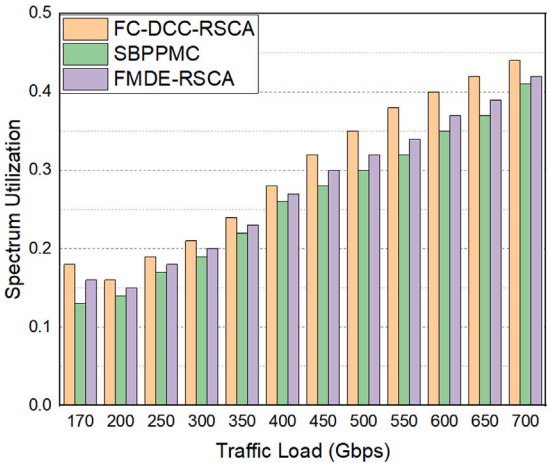


Fig. 9. Spectrum utilization for NSFNET.

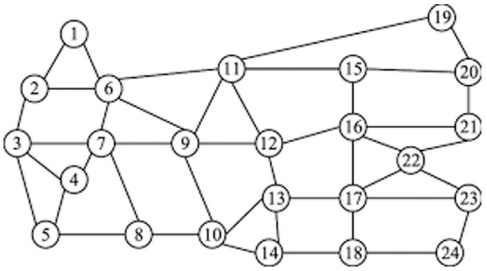


Fig. 10. USNET topology.

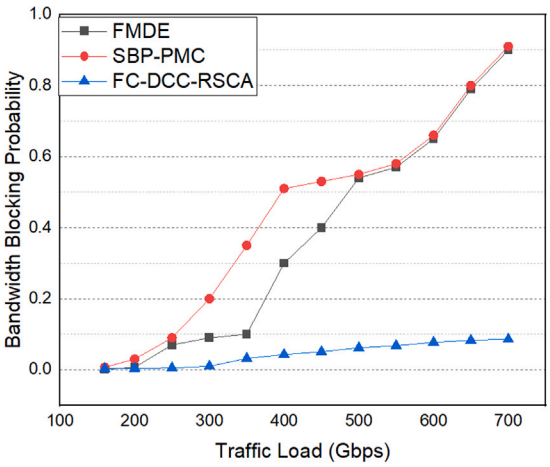


Fig. 11. Bandwidth blocking probability for NSFNET.

The holistic technique employed by USNET, which incorporates considerations of fragmentation coefficient and dynamic core changing, surpasses the performance of SBP-PMC and FMDE algorithms by optimizing resource allocation, efficiently managing spectrum, and dynamically adjusting to network requirements. This holistic optimization strategy minimizes problems caused by fragmentation and effectively allocates resources, leading to enhanced network performance with much fewer occurrences of connection blockage.

USNET offers advanced technology and services that are specifically tailored to satisfy the needs of optical networking. The advantages include fast data transfer, low latency connections, scalability, and support for data-intensive applications such as high-definition video streaming, cloud services, and real-time data processing. Consumers

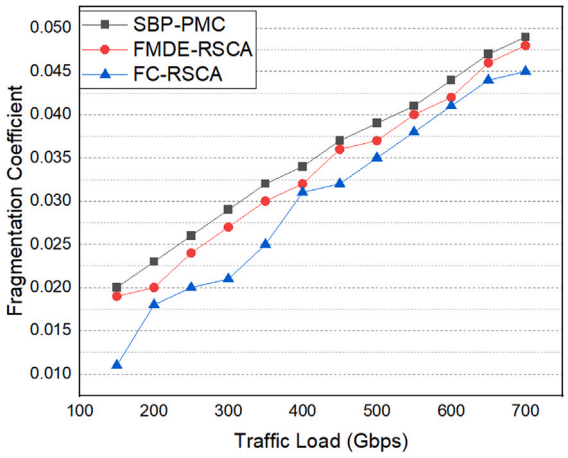


Fig. 12. Fragmentation coefficient for NSFNET.

looking for flexible and affordable solutions will find the network's cost-effectiveness and scalability attractive due to its ability to offer flexible plans that can accommodate various bandwidth requirements. The trends depicted in Fig. 11 have been noticed. As the need for traffic load increases, the BBP of all three algorithms also increases.

Furthermore, the suggested approach exhibits superior performance in BBP compared to existing algorithms such as SBP-PMC and FMDE schemes as shown in Figs. 11 and 14. The reason for this is that both the SBP-PMC and FMDE algorithms employ a survivable multipath strategy, which allows them to minimize the utilization of spectrum resources while ensuring the availability of the requested service. The success of FC-DCC-RSCA in decreasing BBP compared to FMDE and SBP-PMC within USNET stems from its superior management of spectrum fragmentation, its dynamic adaptability in core allocation, its complete optimization strategy, and its more efficient resource management algorithms. The combination of these elements results in a decrease in BBP in FC-DCC-RSCA, emphasizing its superiority compared to FMDE and SBP-PMC algorithms in the setting of the USNET network. Finally, we can show that the BBP of all algorithms in the USNET topology is lower than that of the NSFNET topology. This is because the USNET architecture has more connectivity, and there are more links and spectrum resources in the environment, which makes it easier to discover available resources for requests. FC-DCC-RSCA exhibit BBP of 0.085 and 0.086 at 700 traffic load of gbps for USNET and NSFNET topology as shown in Figs. 14 and 11. This is because the USNET architecture has more connectivity, and there are more links and spectrum resources in the environment, which makes it easier to discover available resources for requests. FC-DCC-RSCA exhibit BBP of 0.085 and 0.086 at 700 traffic load of erlang for USNET and NSFNET topology.

It is evident that the spectrum utilization in USNET is lower when compared to techniques in NSFNET. This is due to the increased number of links in USNET compared to NSFNET. When an equal number of requests are transmitted, the distribution of requests to each link in the USNET network topology results in a lower load, and the link has a greater amount of unused spectrum resources. Consequently, when sending an equal number of requests, the efficiency of spectrum usage in USNET is worse than that in NSFNET. FC-DCC-RSCA exhibit SU of 0.42 and 0.44 for a USNET and NSFNET topology at 700 Erlang as shown in Figs. 9 and 13. Dynamic core-changing techniques based on fragmentation coefficients provide a decreased fragmentation ratio due to their targeted optimization strategies in spectrum allocation and dynamic resource management. The purpose of these strategies is to reduce fragmentation by effectively allocating spectrum resources and guaranteeing the availability of uninterrupted frequency blocks for allocation. The incorporation of dynamic core-changing capabilities

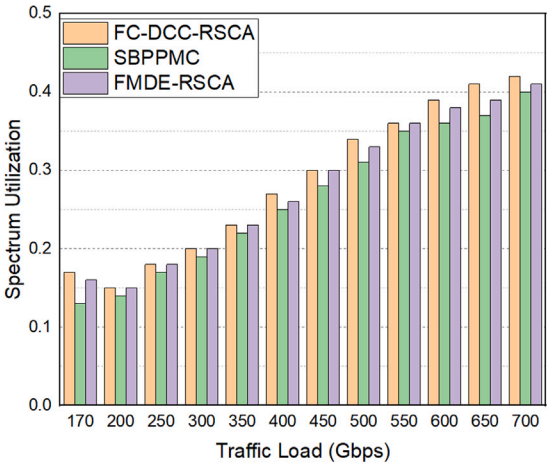


Fig. 13. Spectrum utilization for USNET.

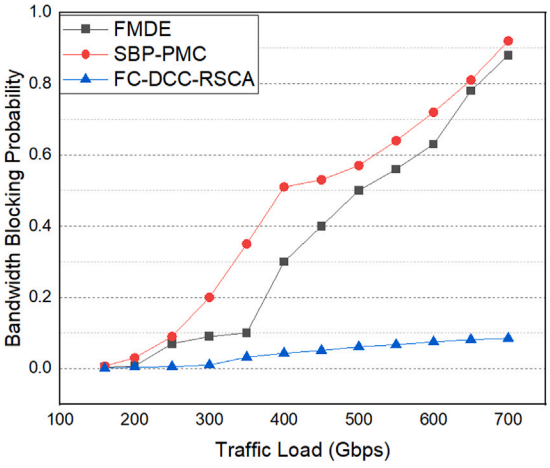


Fig. 14. Bandwidth blocking probability for USNET.

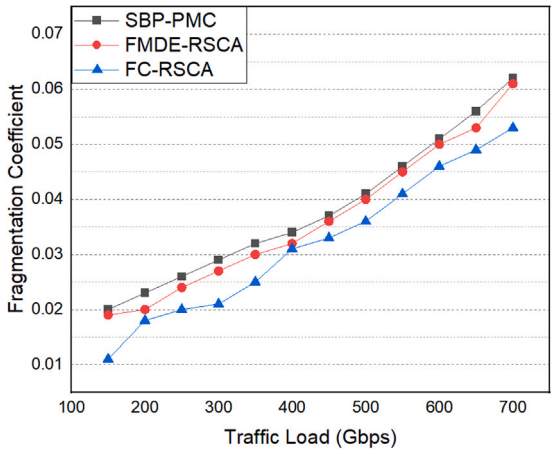


Fig. 15. Fragmentation coefficient for USNET.

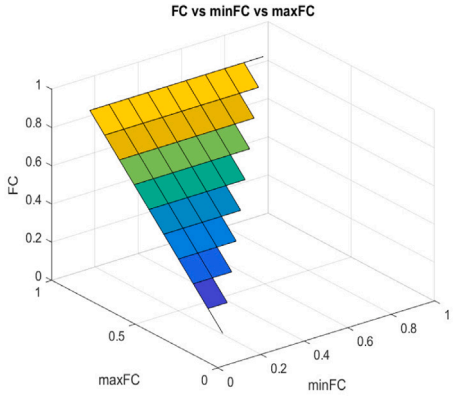


Fig. 16. 3D representation of minFC, maxFC and FC.

enables the network to make adaptive modifications to core allocations, resulting in effective resource utilization and allocation that minimizes fragmentation. Furthermore, the focus on effective resource allocation is of utmost importance. These strategies have the ability to adjust and respond to changing network demands and traffic patterns. They optimize the distribution of spectrum and core resources to reduce fragmentation. Through the implementation of complete optimization algorithms that involve spectrum allocation, core utilization, and routing considerations, these technologies successfully decrease fragmentation and maintain a lower fragmentation ratio. In summary, combining fragmentation coefficient-based methodologies with dynamic core-changing features allows for optimal resource utilization, resulting in a reduced fragmentation ratio in the network architecture. FC achieved around 0.053 around for FC-DCC-RSCA scheme lesser than other SBP-PMC and FMDE techniques as shown in and Figs. 12 and 15.

The combined method takes into account both route and core selection, allowing for a more delicate fragmentation control strategy. It computes fragmentation coefficients for each possible combination, guaranteeing that the chosen core makes the best use of available spectrum resources. Other techniques, on the other hand, are primarily concerned with path-based fragmentation minimization. The integrated algorithm, which incorporates core-level consideration, extends this and results in a more thorough approach to fragmentation reduction. In addition, the fragmentation measurement for the proposed algorithm is more accurate than others. Consequently, the FC-DCC-RSCA algorithm yields superior performance results in comparison to other algorithms. FC-DCC-RSCA has a BBP of 0.051 in the NSFNET network when the traffic load is 450 Erlang, compared to SBP-PMC and FMDE-RSCA.

The performance of our proposed FC-DCC-RSCA in terms of SU is also incorporated into the NSFNET technology. Under every load condition, the SU is greater than those of the two other comparison algorithms. On the one hand, this is due to the fact that the algorithm employs adaptive core allocation. By permitting dynamic core modification, the system is able to adapt to fluctuating traffic demands and network conditions. This allows the algorithm to select cores that are better aligned with the intended spectrum, decreasing the likelihood of spectrum being wasted. FC identifies routes and cores for which the available spectrum coordinates well with the necessary contiguous slots. This reduces the degree of spectrum fragmentation, allowing for more efficient use of available resources. In addition, FC helps allocate spectrum slots for a particular communication, thereby reducing spectrum waste. On the other hand, SBP-PMC is only based on shared paths and displays the lowest spectrum utilization. FC-RSCA therefore has greater spectrum utilization than other algorithms. For the traffic load of 450 Erlang, the spectrum utilization of the FC-DCC-RSCA increases by 12.5% and 6.25%, respectively, as compared to SBP-PMC and FMDE.

Among the two other schemes that take fragmentation into account, we compare their fragmentation ratios in the NSFNET topology as shown in Figs. 12 and 15.

The fragmentation ratio for each scheme increases with the traffic volume, as shown in the figure. Compared to the other two comparison algorithms, the FC-DCC-RSCA algorithm proposed in this paper

demonstrates the lowest network fragmentation ratio. Furthermore, the process of resource selection and allocation consistently takes into account the potential influence of fragmentation on the availability of requested resources. It strives to allocate a greater quantity of larger, unutilized spectrum blocks for future requests. Additionally, it is important to acknowledge that this phenomenon also demonstrates the fundamental cause of superiority in terms of BBP and resource utilization. This outcome is anticipated since fragmentation directly impacts the blocking probability of requests. In the NSFNET network, it has been observed that when the Erlang value is set to 450, the FC-DCC-RSCA algorithm exhibits a reduction in its FR by 13.5% and 2.7% when compared to other algorithms. A three-dimensional representation that depicts the concepts of minFC, maxFC, and FC within the context of spectrum allocation might serve as a valuable visualization tool. This analysis can facilitate the visualization of the interplay between these elements and enhance comprehension of their influence on the allocation procedure. The following is a suggested approach for interpreting a given Fig. 16.

Axes: This axis perhaps represents the spectrum allocation combinations or pairings. Each x-axis point could correspond to a unique combination of path, core, source, and destination (k, c, s, and d). This axis represents the FC (fragmentation coefficient) values. It indicates the degree of fragmentation for each spectrum allocation pair.z-axis: The z-coordinate on the graph corresponds to a specific value of minFC and maxFC as shown in Fig. 16

Surface Plots: A plane or surface that goes across the x–y plane would be the surface plot for minFC. It shows the lowest amount of fragmentation that is allowed for each spectrum allocation pair. This area is most likely going to be flat on the x–y plane.

maxFC Surface Plot: The surface plot for maxFC would be a different plane or surface, maybe even one that runs parallel to the x–y plane. For each spectrum allocation pair, it shows the highest amount of fragmentation that is allowed (maxFC).

Intersection of minFC and maxFC Surfaces: In 3D space, the area where these two surfaces meet is called a confined volume or area. This bounded area shows the fragmentation amounts that are allowed based on minFC and maxFC for the spectrum distribution pairs.

FC Points in 3D Space: In 3D space, the FC numbers with each band allocation pair are shown as points. Based on whether the FC number is below or above the minFC surface, these points will be above or below the x–y plane. The position of each FC point in relation to the minFC and maxFC surfaces indicates that the spectrum allocation pair meets the fragmentation constraints. If a point is within the bounded region defined by minFC and maxFC, it means that the fragmentation is within acceptable limits. This visualization can help it quickly and easily decide which spectrum allocation pair is best.

By initially increasing the $XT_{threshold}$, a greater number of channels can be accommodated within each core without experiencing substantial cross-talk interference. Consequently, there is a rise in the utilization of the spectrum. Beyond a certain threshold, the incremental increase in the XT threshold may not yield a substantial improvement in spectrum utilization. This phenomenon can occur due to the system already functioning optimally, whereby increasing the threshold does not yield significant advantages. Adopting a flexible approach to modifying core values at each threshold is a commendable strategy. This enables the observation of how different cores react to changes in the $XT_{threshold}$. This holds particular significance in multicore systems, as the actions of each individual core can have a direct impact on the overall performance of the system.

6. Conclusion and future work

This research introduces a unique paradigm for spectrum allocation that aims to effectively manage light path demands in optical networks. The algorithm being presented utilizes a heuristic methodology that incorporates path selection and XT constraint analysis in order to optimize the allocation of spectrum resources while simultaneously minimizing both the chance of bandwidth blockage and fragmentation. Next, we assess FC using the CASR model, where we use a filtering and prioritization process to identify the pairs that fall between the predetermined minFC and maxFC limits. The flexibility of our algorithm to adapt to changing network circumstances and fluctuating demand patterns is a significant advantage. By using dynamic allocation of processing cores and looking at different routing options, our solution shows that it can work even when the amount of traffic changes. The FC-DCC-RSCA demonstrates an improvement in spectrum utilization by 12.5% and 6.25%, along with a reduction in its FR by 13.5% and 2.7%. Additionally, it displays a BBP of 0.051 for a traffic load of 450 Erlang when compared to SBP-PMC and FMDE-RSCA. Future research endeavors may focus on conducting further enhancements to the algorithm, taking into account supplementary variables such as energy efficiency and scalability. This will enable the continuous progression of the current state-of-the-art in the realm of optical network management.

CRediT authorship contribution statement

Vasundhara: Writing – original draft, Validation, Software, Methodology, Conceptualization. **Abhilash Mandloi:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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